

Topic 8.1
Support Vector Machines

Introduction to Support Vector Machines

By Dr Lee How Chinh

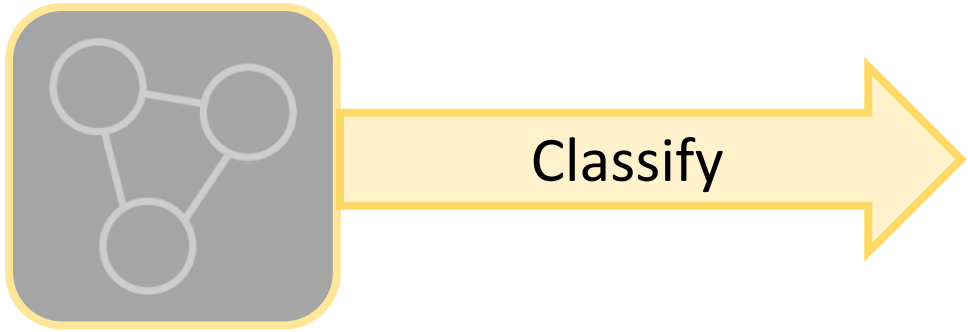
Objectives

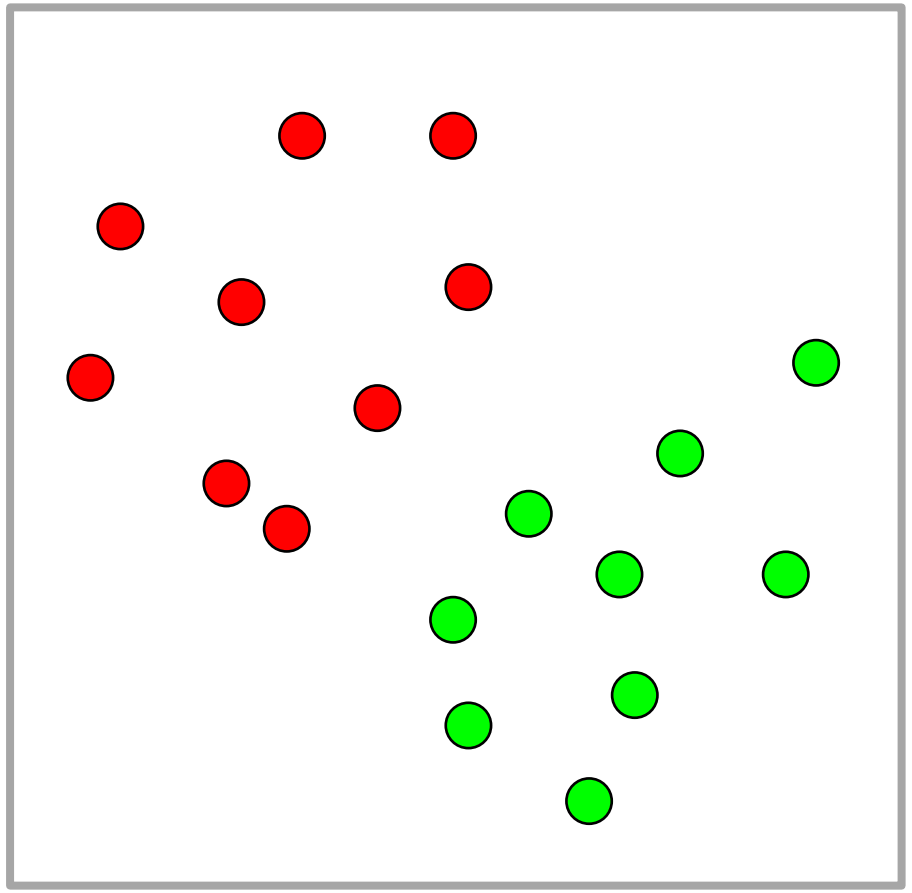
- Describe what is Maximum-Margin Hyperplane and Support Vectors.

Introduction to Support Vector Machines

Support Vector Machines (SVMs)

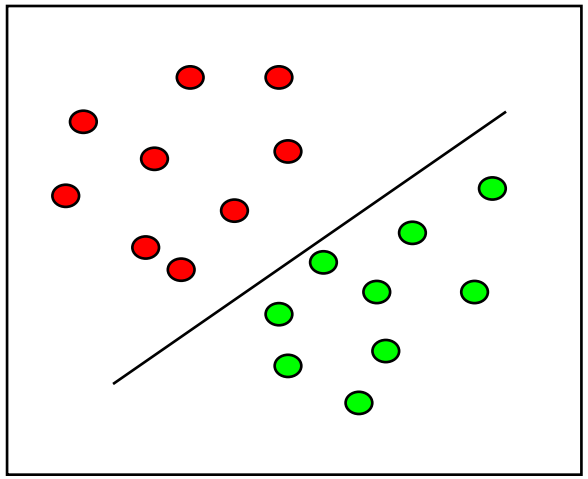
Support Vector Machine

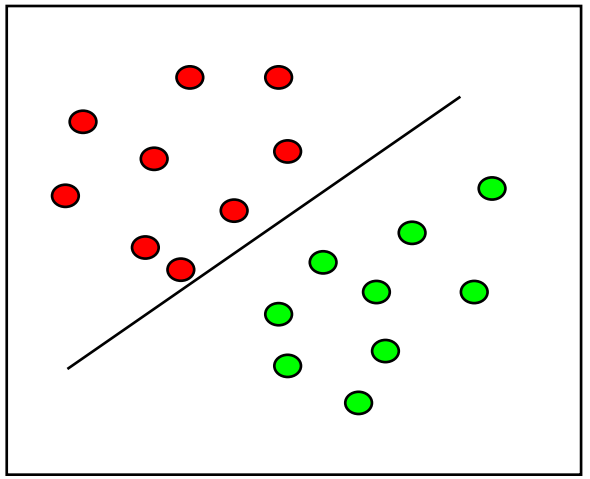


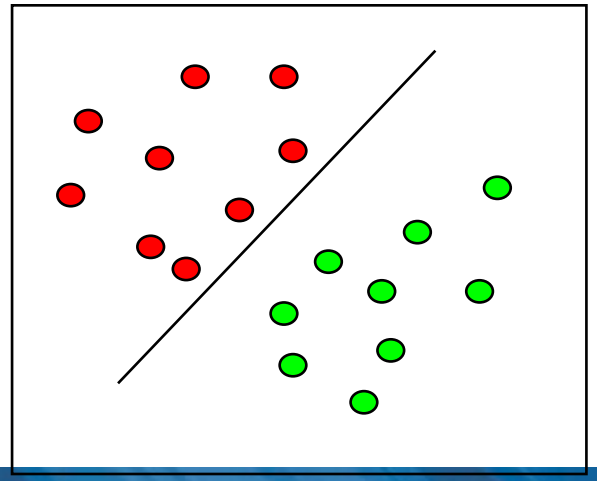


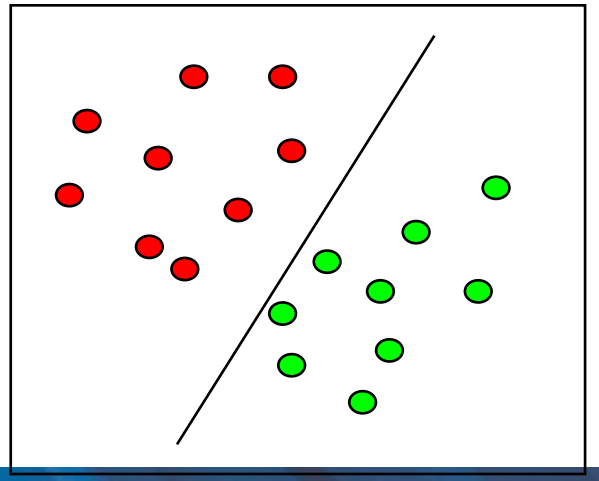
- decision
- rank
- estimate

How Do You Classify Red versus Green?







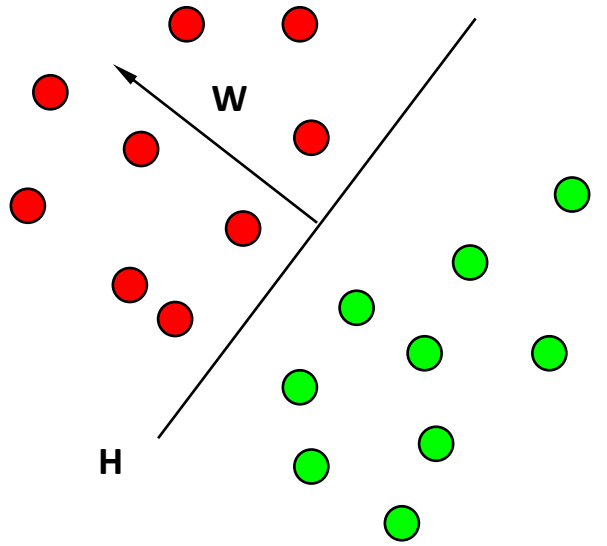


Linear Separation of the Training Data

- A separating hyperplane H is given by the following:
  - the normal vector w
  - an additional parameter, b, called bias

$$H = \{\langle w, x \rangle + b = 0\}$$

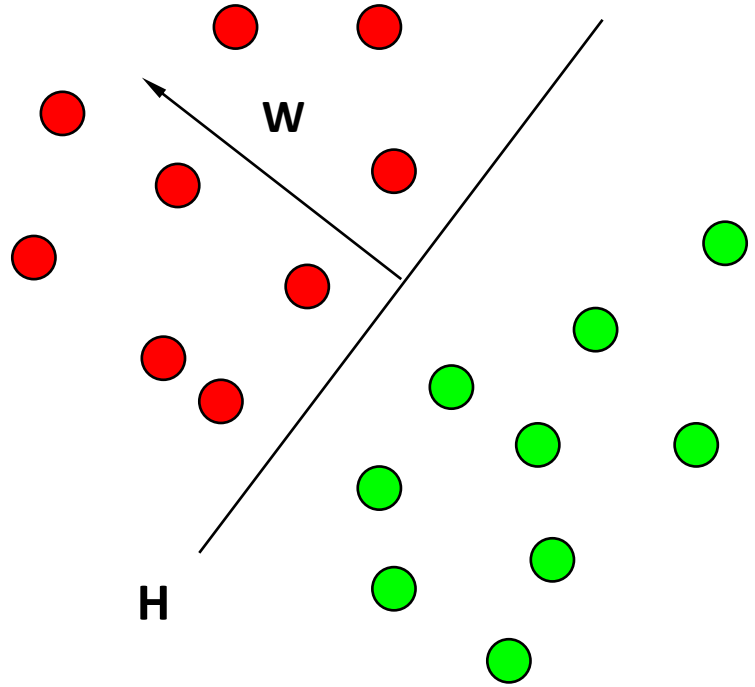
Dot product



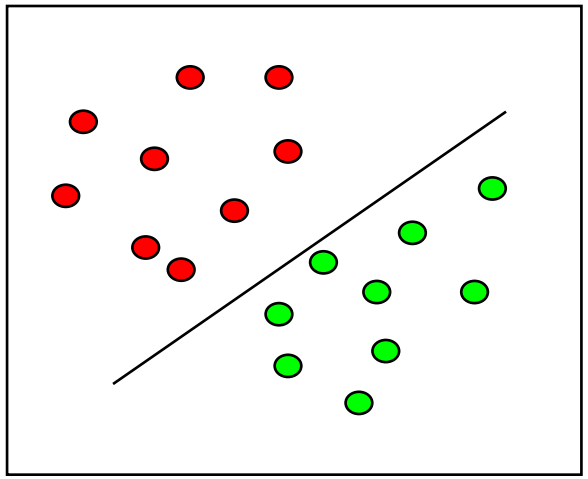
Training versus Prediction

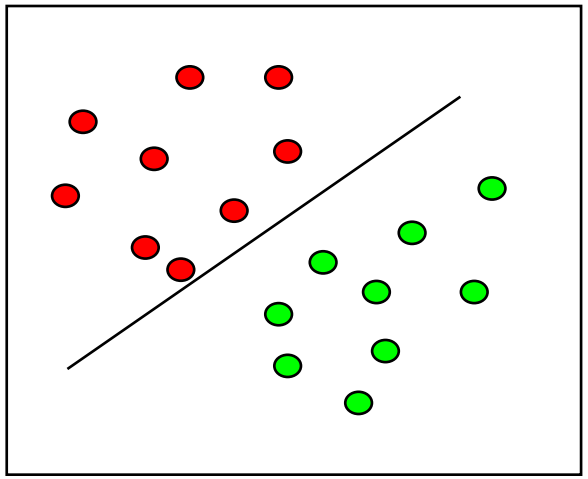
- Training:
- Select w and b in such a way that the hyperplane separates the training data – that is, the construction of a hyperplane.

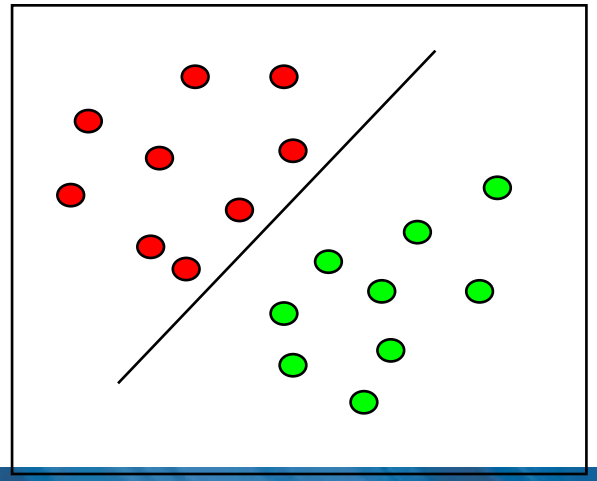
- Prediction of the class for a new observation:
  On which side of the hyperplane (+/-) is the new data point located?

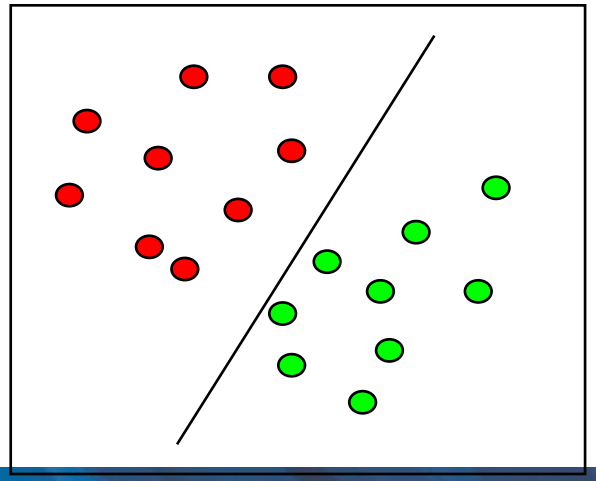


Which Hyperplane Is the Best One?

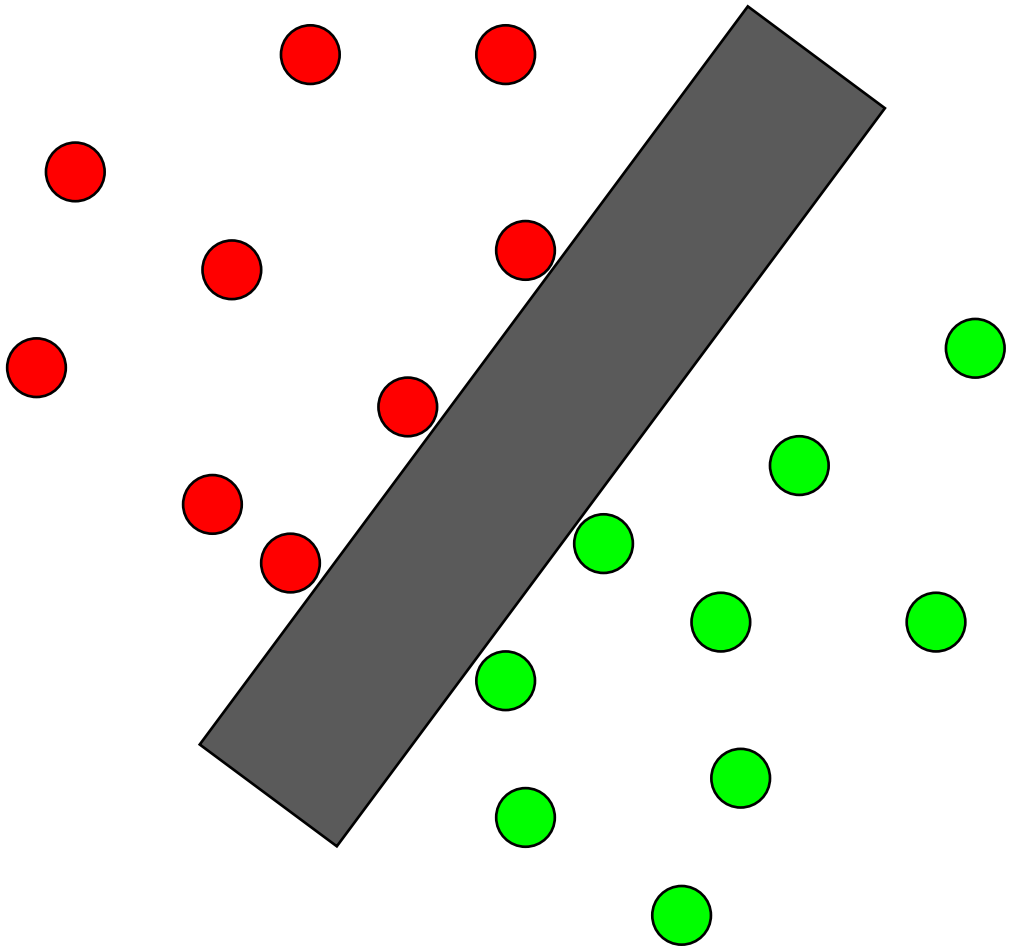




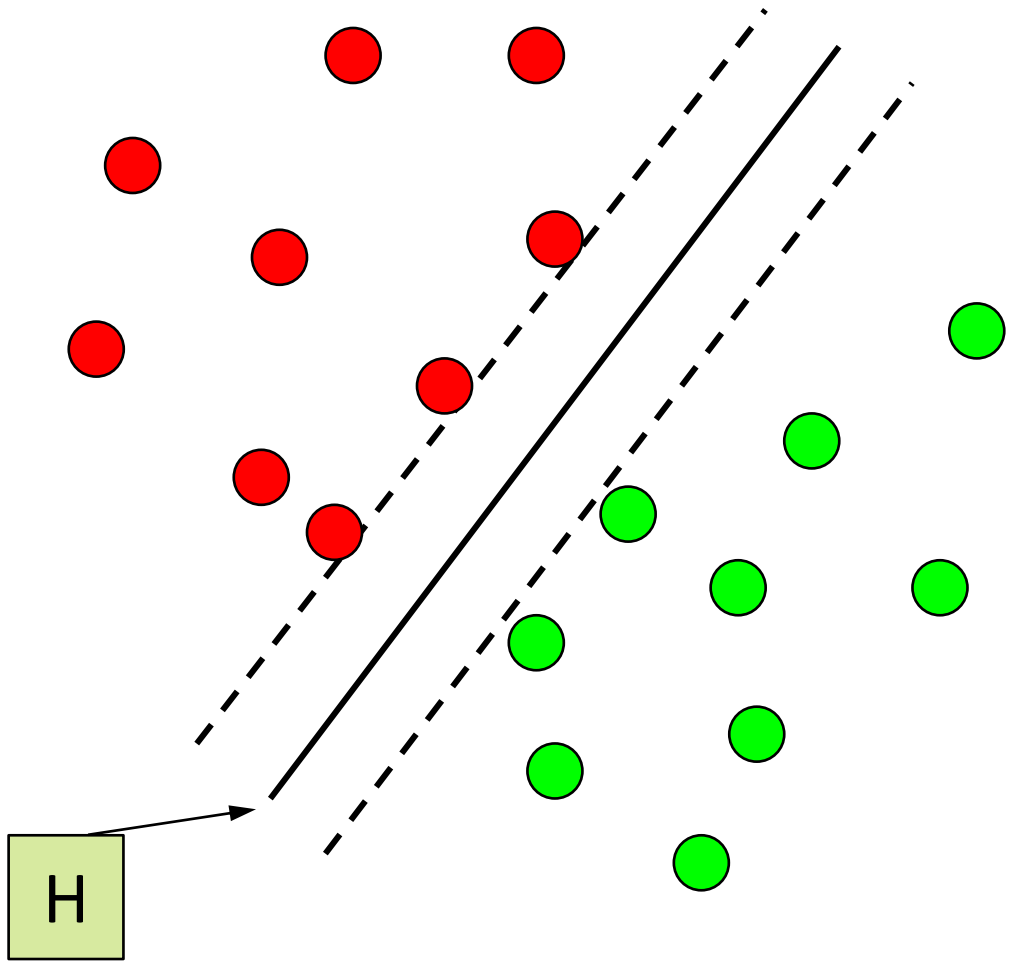




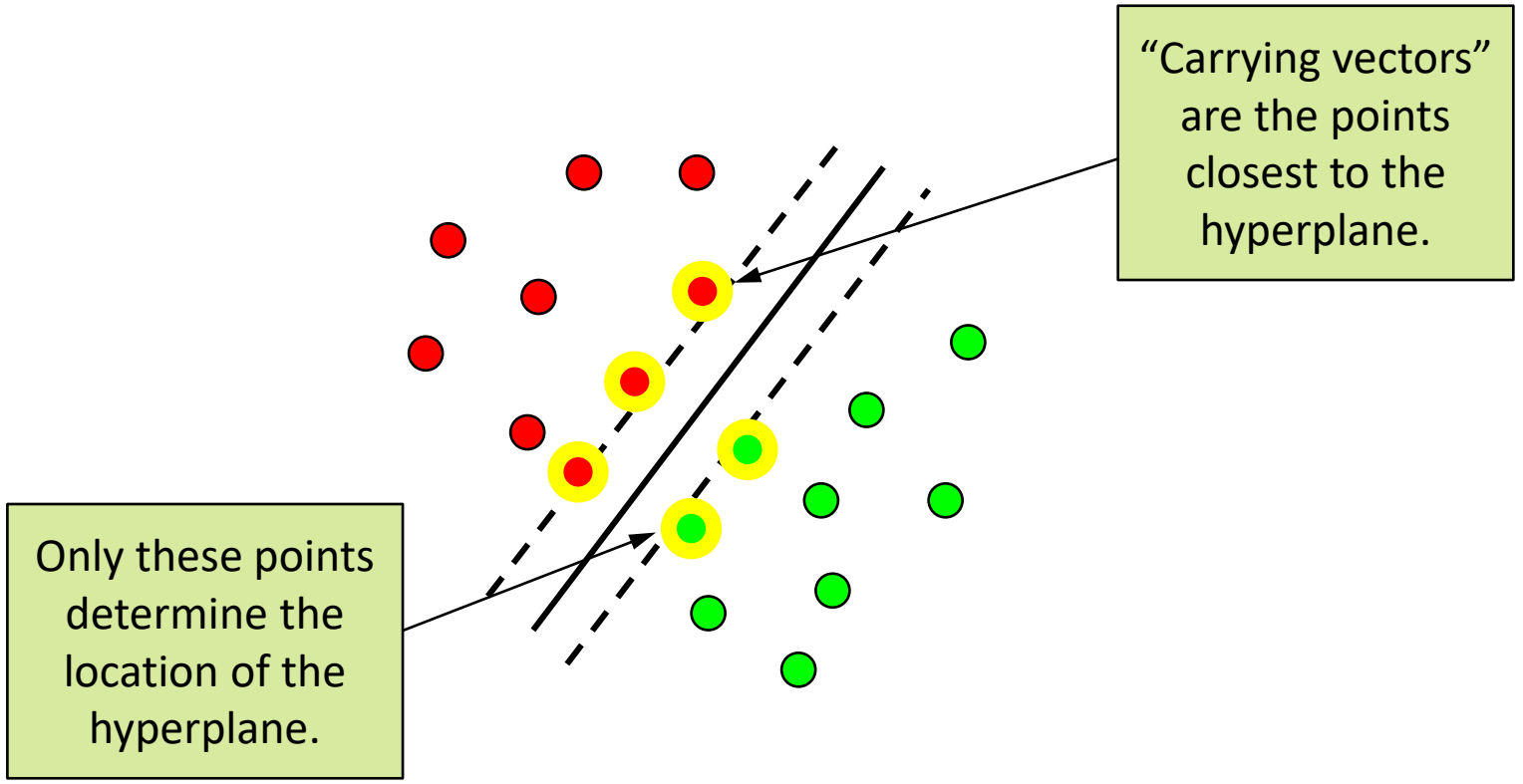
A “Fat” Hyperplane



A Maximum-Margin Hyperplane



What Are the Support Vectors?



Lesson Summary

- Support Vector Machines are good for classifying observations and producing decision-type predictions.
- A separating hyperplane H is defined by a normal vector *w* and an additional parameter *b*, called *bias*. $H = \{\langle w, x \rangle + b = 0\}$
- The best separating hyperplane is the Maximum-Margin Hyperplane.
- Support Vectors are the points closest to the hyperplane and determine the location of the hyperplane. Therefore, SVMs don't face the curse of dimensionality.

How does a SVM optimise the complexity?



MONASH University



Topic 8.2
Support Vector Machines

Optimising the complexity of the SVM

By Dr Lee How Chinh

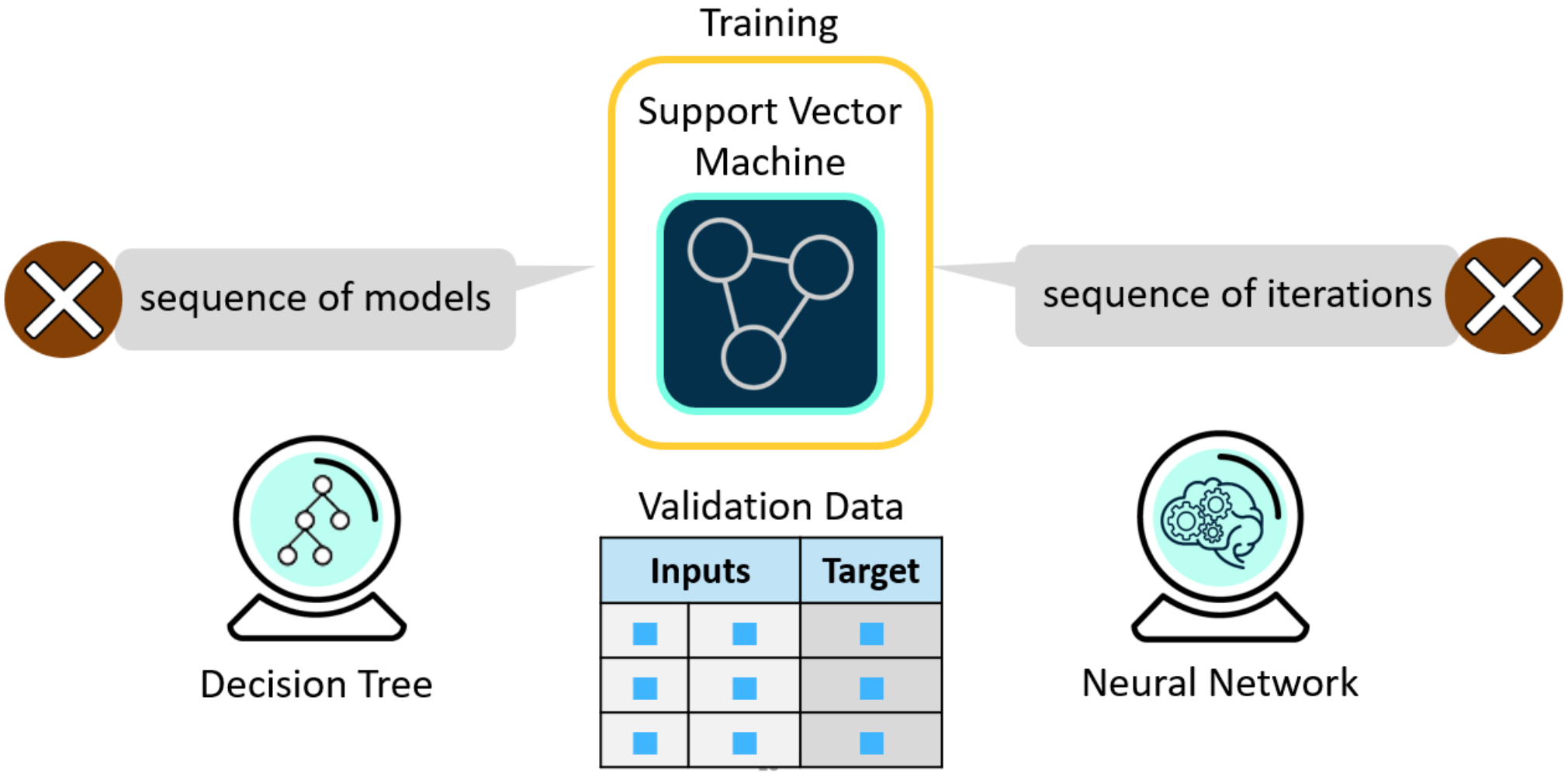


Objectives

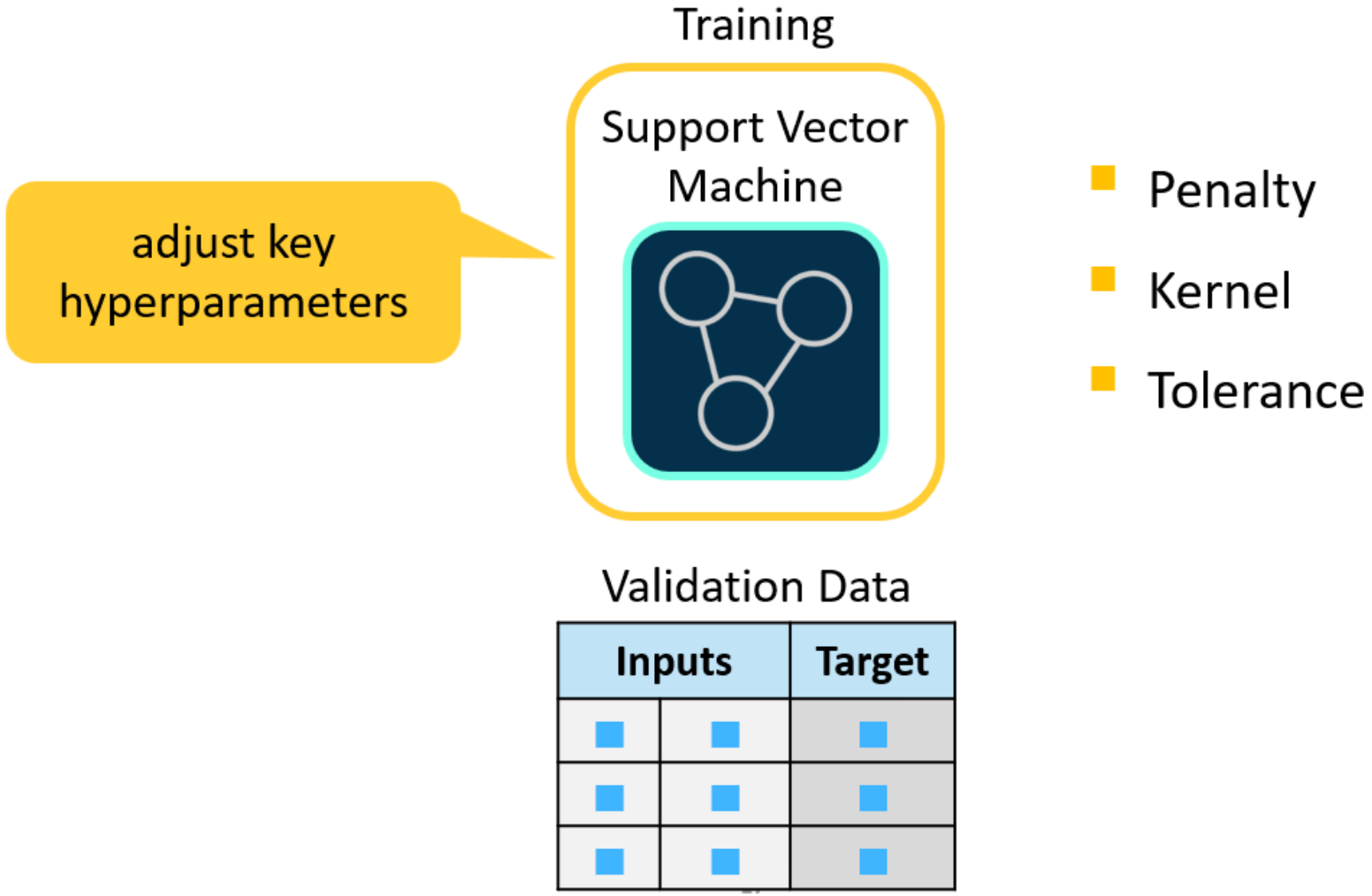
- Describe and explain the optimising complexity by implementing the soft margin hyperplane.

Optimising the complexity of the SVM

Improving the Support Vector Machine Model



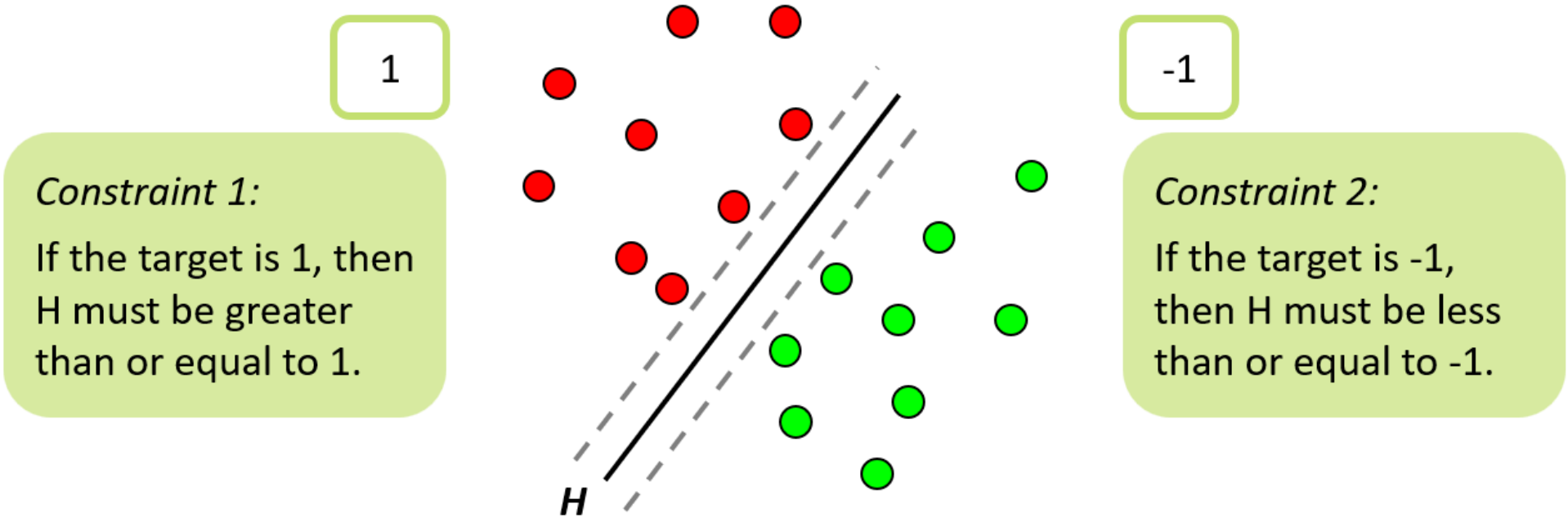
Improving the Support Vector Machine Model



- Penalty
- Kernel
- Tolerance

Optimisation Problem

The solution for finding the separating hyperplane H becomes an *optimisation* problem under two constraints:

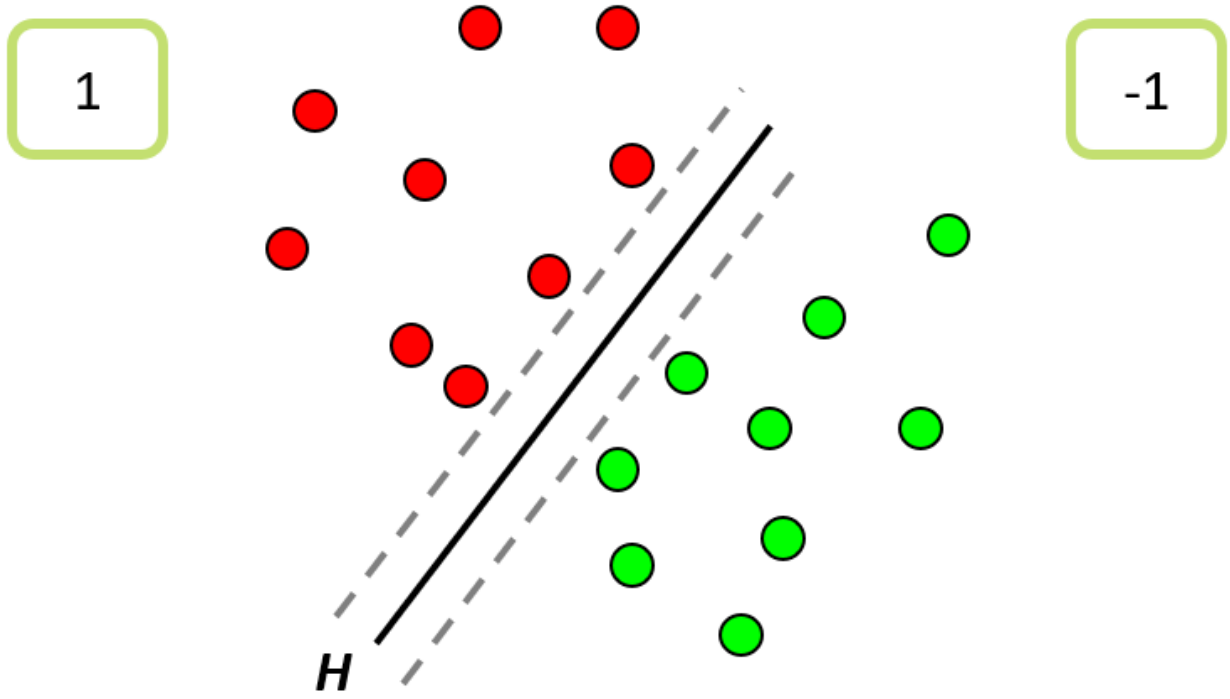


The binary target is written as +/- 1 for mathematical convenience.

Optimisation Problem

The two constraints can be combined into a single constraint:

The product of the target and H must be greater than or equal to 1 for all cases.



Optimisation Problem

- Mathematically, these constraints are written as follows:

$$\langle w, x_i \rangle + b \geq 1 \quad \text{if } y_i = 1 \text{ and}$$
$$\langle w, x_i \rangle + b \leq -1 \quad \text{if } y_i = -1$$

- However, these two constraints can be combined into a single constraint where

$$y_i \cdot (\langle w, x_i \rangle + b) \geq 1 \quad \text{for } i = 1, 2, 3, \ldots, n.$$

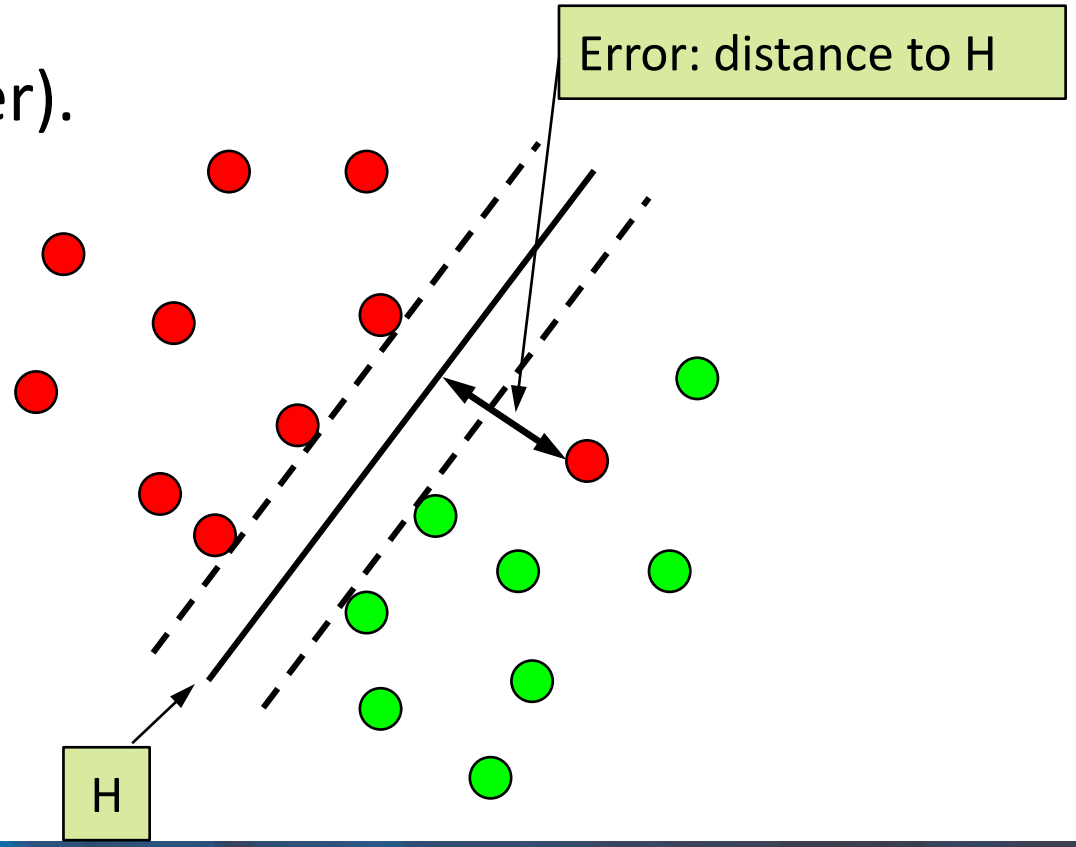
Training Data Not Linearly Separable

Errors are accounted for during the optimisation process:

Penalty: C*(distance to hyperplane)

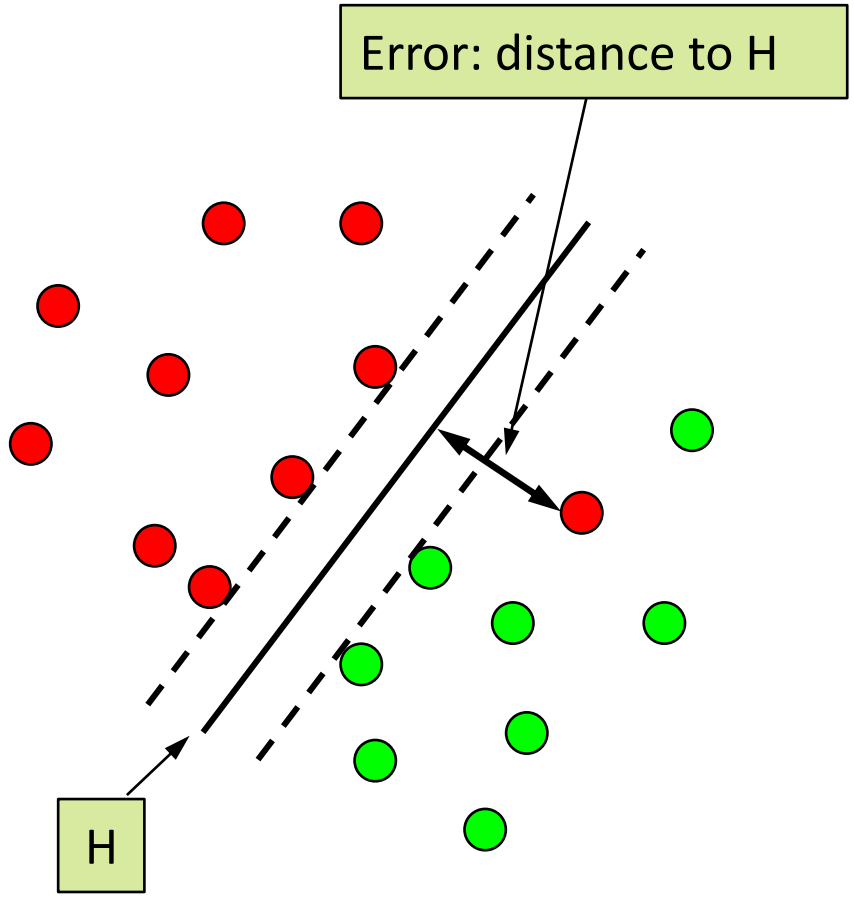
C is an error weight (regularization parameter).

We have a so-called **soft margin hyperplane** if the data points are not linearly separable. In this case, we need to account for errors that the separating hyperplane might make. During optimisation, the distance between a point in error and the hyperplane is typically denoted by ξ.



Training Data Not Linearly Separable

- For large values of C, the optimization chooses a smaller-margin hyperplane if that hyperplane does a better job of getting all the training points classified correctly.
- Conversely, a very small value of C causes the optimizer to look for a larger-margin separating hyperplane, even if that hyperplane misclassifies more points. For very tiny values of C, you should get misclassified examples, often even if your training data is linearly separable.



Lesson Summary

- The SVM model is improved by adjusting key hyperparameters of penalty, kernel and tolerance.
- The targets are labelled as -1 and +1, so $y_i \cdot H \geq 1$.
- Soft-margin hyperplane is referred to when the data points are not linearly separated.
- Penalty is defined by $C \cdot \xi$, and the optimization problem is solved by minimizing $\|w\|^2 + C \cdot \sum \xi_i$, where ξ is the distance between error point and hyperplane.

How about the data points that can’t be separated by a linear hyperplane?



MONASH University



Topic 8.3
Support Vector Machines

Nonlinear Classifier: Kernel Trick

By Dr Lee How Chinh



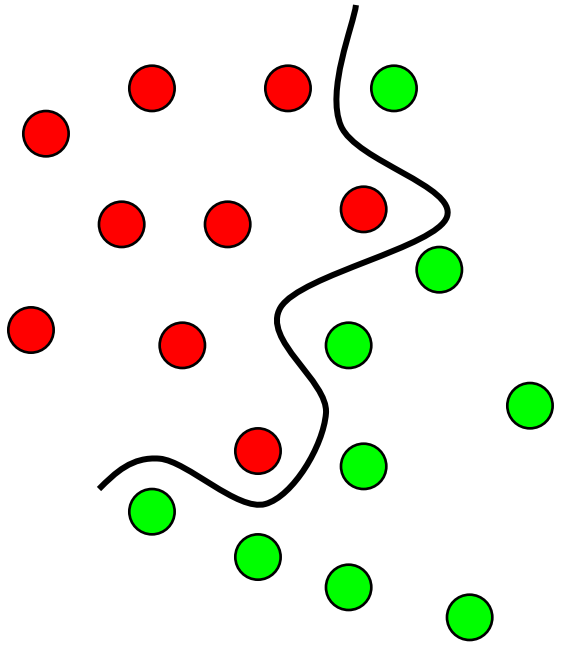
Objectives

- Explain the idea of feature space.
- Use kernel trick to construct a feature space.

Feature Space

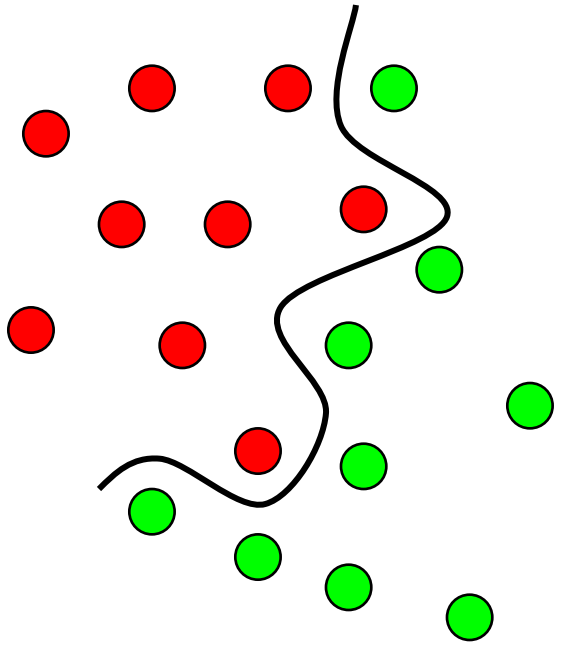
Problem: Not Linearly Separable Data Points

Input space 2-D



Problem: Not Linearly Separable Data Points

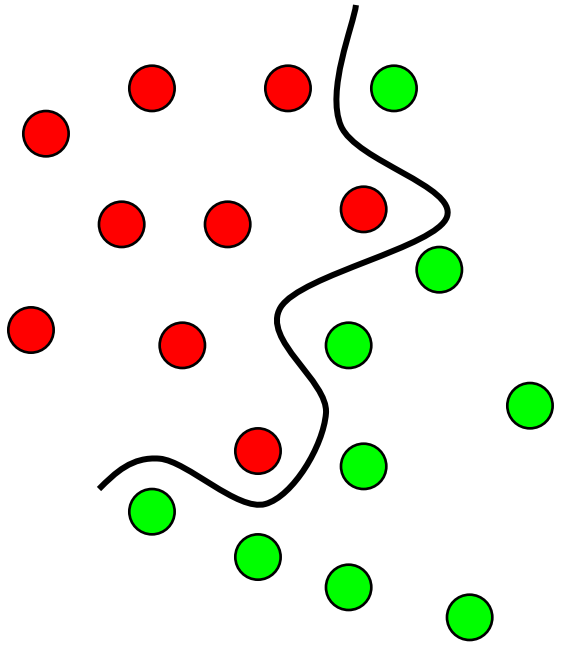
Input space 2-D



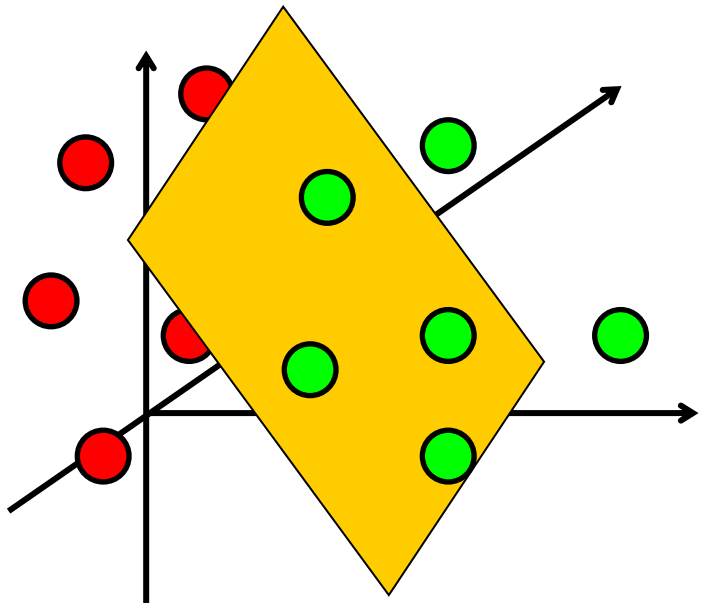
- Feature space is a nonlinear transformation of the input variables into a high-dimensional feature space.
- The maximum-margin hyperplane is constructed in the high-dimensional feature space.

Problem: Not Linearly Separable Data Points

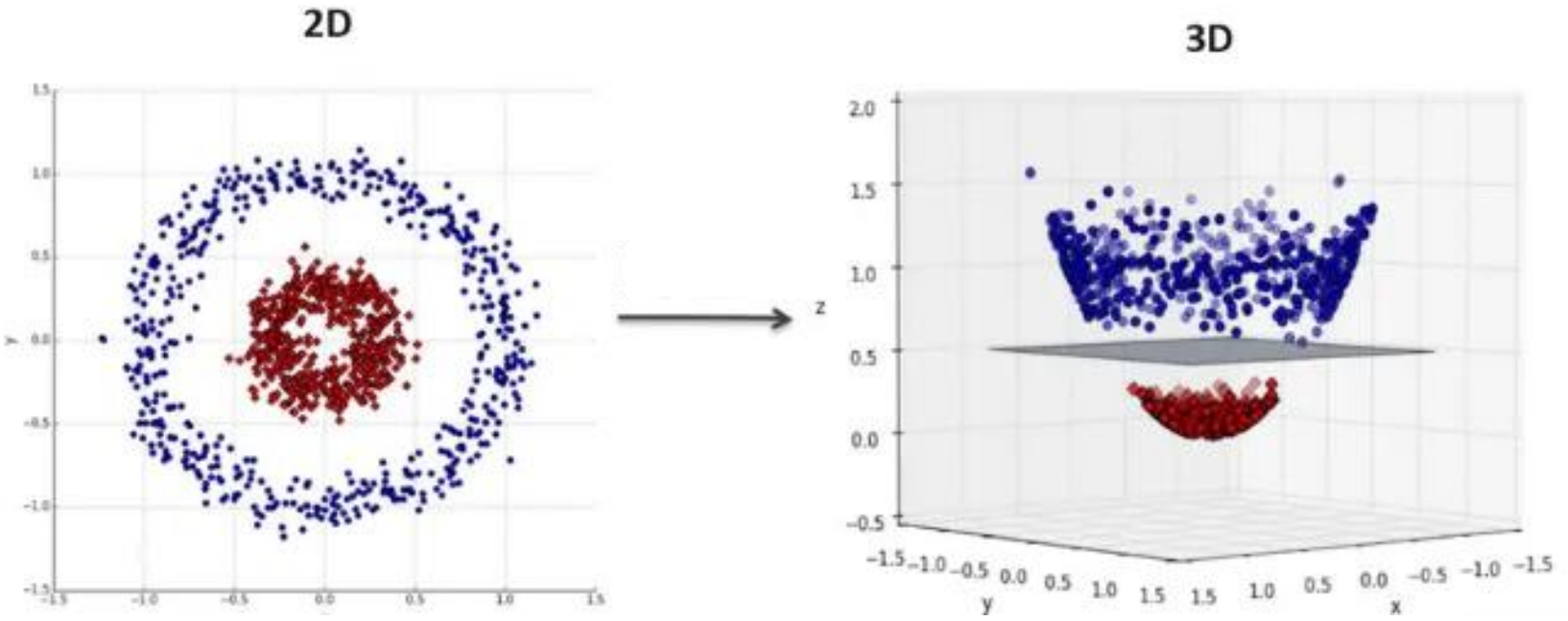
Input space 2-D



Feature space 3-D



Problem: Not Linearly Separable Data Points



Kernel Trick

Important Observations

- Estimating the parameters for a support vector machine and classifying a new case both require dot products of the form:

$$\langle x_i, x_j \rangle$$

- The solution in a higher dimension requires dot products on transformed data:

$$\langle \Phi(x_i), \Phi(x_j) \rangle$$

- **Problem:**
  Dot Products of the form $\langle \Phi(x_i), \Phi(x_j) \rangle$ are difficult to calculate.

Solution: The Kernel Trick

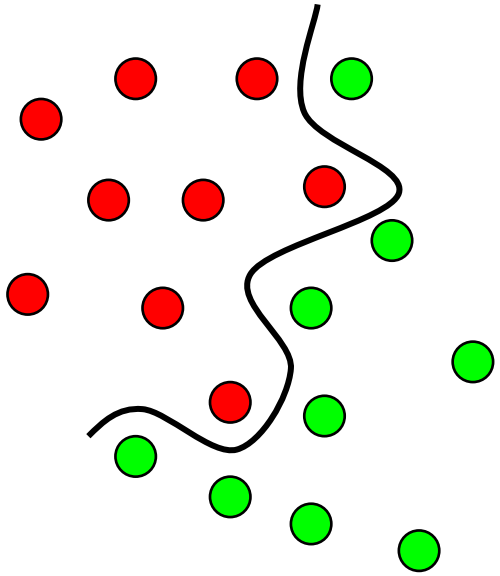
- We use a kernel function, living in the input space, but behaving as a dot product in the feature space:

$$K(x_i, x_j) = \langle \Phi(x_i), \Phi(x_j) \rangle$$

- Trick: We do not have to know $\Phi(x)$ explicitly!

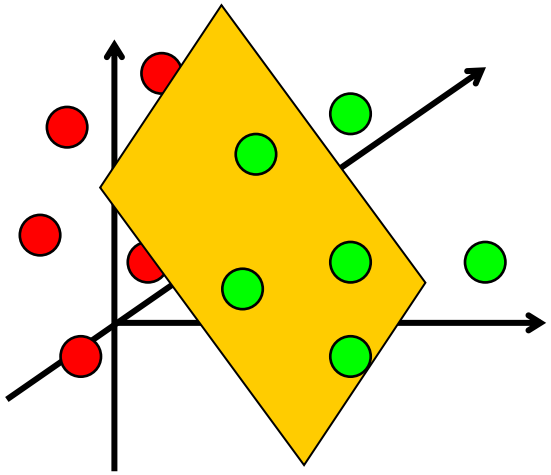
Solution: The Kernel Trick

Input space



Nonlinear separation
with kernel function

Feature space



Linear separation
with dot product

Examples of Kernel Functions

- Linear:

$$\mathcal{K}(x_i, x_j) = \langle x_i, x_j \rangle$$

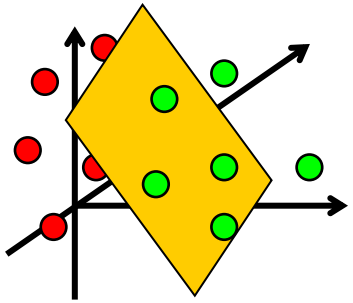
- Polynomial:

$$\mathcal{K}(x_i, x_j) = \left(\langle x_i, x_j \rangle\right)^d$$

Parameters for SVMs for Classification

- The kernel function and its parameters
  - Avoids the explicit mapping that is needed to get linear learning algorithms to learn a nonlinear function or decision boundary.
- The penalty C (regularisation term)
  - Balances model complexity and training error. A larger penalty value creates a more robust model at the risk of overfitting the training data.
- The tolerance
  - Balances the number of support vectors and model accuracy. A tolerance value that is too large creates too few support vectors, and a too small value overfits the training data.

Model Interpretability



A support vector machine has a geometric interpretation.



A support vector machine is a black box.





Lesson Summary

- When the data is nonlinearly separated, a maximum-margin hyperplane is built in a high-dimensional feature space. Where the feature space is the nonlinear transformation of the input variables.
- Estimating the SVM parameters using the dot products on the nonlinear transformed data of the feature space is challenging.
- The kernel function of the inputs can be used to replace the dot product in the feature space without knowing the transformed function.
- In the Model studio, the SVM node provides linear, first and second degree polynomial Kernel functions.
- The SVM models can be optimised by customising the kernel function parameters, the penalty regularisation term, and the tolerance.

How to construct SVM models in SAS Viya?



MONASH University